آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre SSC-I General Mathematics Annual Examinations 2025

Introduction

This document has been prepared for the teachers and candidates of Secondary School Certificate (SSC) Part I (Class IX) General Mathematics. It contains comments on candidates' responses to the 2025 SSC-I Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used. It is imperative to refer to command word guide available on AKU-EB website for understanding the expectations of the command word.

General Observations

Candidates performed really well in some concepts, such as, sets, matrices and degree conversion through graph. However, candidates who did not score well mostly failed to understand the demands of the questions, often misinterpreting the command words and the stimuli.

Mentioned below are few concepts that teachers need to focus so that the candidates may perform better.

- Percentages
- Logarithm

Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

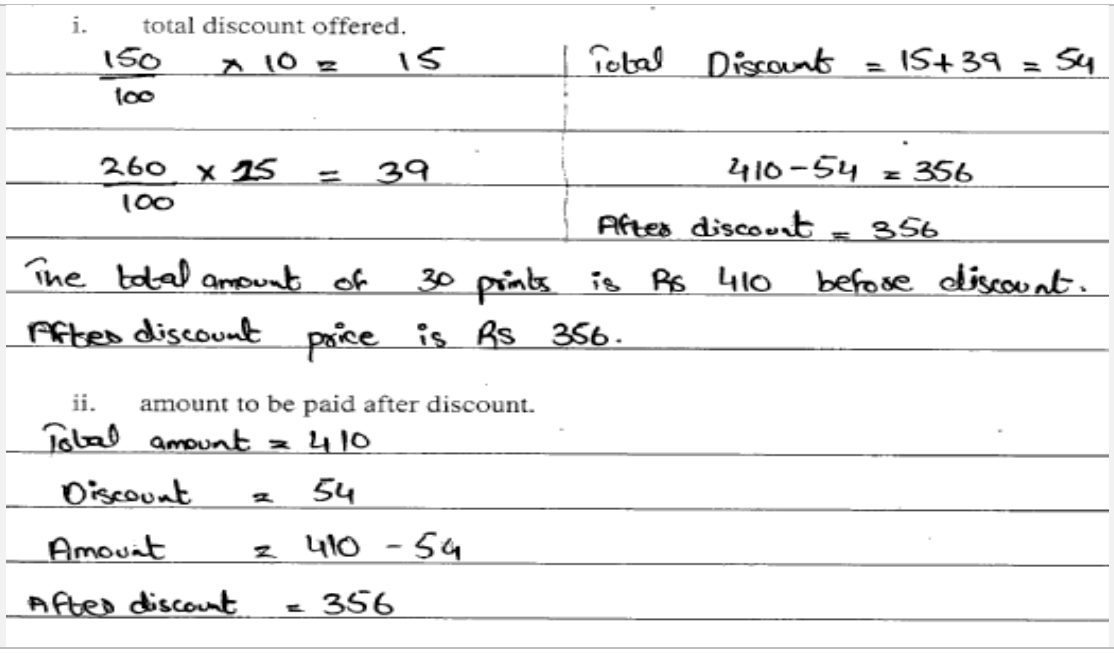
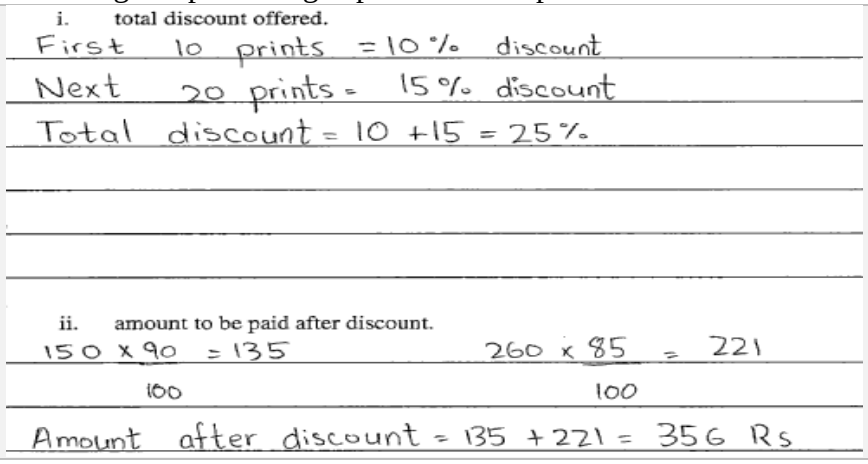
DETAILED COMMENTS

Constructed Response Questions (CRQs)

Question No. 1a

Candidates were given the choice to attempt any ONE out of the two questions: 1a and 1b.

Question Text	AB Studio offers the following discounts as tabulated below. **Quantity** / **Price Before Discount (Rs)** / **Discount Percentage** **First 10 Prints** / 150 / 10% **Next 20 Prints** / 260 / 15% If a client requests for 30 prints, then calculate the following: i. total discount offered. ii. amount to be paid after discount.
SLO No.	1.2.2
SLO Text	Solve problems involving percentages.
Max Marks	3
Cognitive Level	A*
Checking Hints	i. 1 mark for calculating price before discount 1 mark for finding the total discount offered ii. 1 mark for finding the amount payable
Overall Performance	This question aimed to assess candidates' understanding of percentage, discount, and discounted price. Few candidates demonstrated a good grasp of percentage concepts and calculated discounts accurately. However, most candidates struggled with finding the correct discount when applied to two separate values.
Description of Better Responses	In *better responses*, candidates demonstrated a clear and accurate understanding of calculating percentages on different values. They correctly identified and applied the formulae for percentage, added both the amounts to find the correct total discount, and then found amount to be paid after discount as 410 - 54 = 356. Their workings showed logical progression and reflecting a well-developed conceptual understanding of the topic.

Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates demonstrated a limited understanding of calculating discounts on separate values. Instead of applying the correct method, some candidates incorrectly added the discount percentages. Although many candidates managed to compute individual percentages correctly, their overall approach lacked clarity and accuracy, reflecting a superficial grasp of the concept.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy** Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach)	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

• Utilise the resource guide for additional materials	• Practical Demonstration ** For description of each Pedagogy, refer to Annexure A	

Any Additional Suggestion:
Teachers should incorporate focused practice on calculating successive and separate discounts using step-by-step examples and guided comparisons. Emphasis should be placed on the correct application of percentage operations in sequence, rather than treating them additively. The use of contextual word problems and interactive tools such as percentage calculators or discount tables can also enhance conceptual clarity and help candidates distinguish between different discount scenarios more confidently.

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

Question No. 1b

Candidates were given the choice to attempt any ONE out of the two questions: 1a and 1b.

Question Text	The property inherited by Jamal was distributed among his widow, two daughters and a son. If widow received Rs 300,000, then calculate the (**Note**: The share of a son is two times that of a daughter.) i. total worth of the property. ii. share of each daughter.
SLO No.	2.3.2
SLO Text	Solve word problems based on inheritance.
Max Marks	3
Cognitive Level	A
Checking Hints	i. 1 mark for finding the worth of the property ii. 1 mark for finding the remaining amount after the deduction in widow's amount 1 mark for finding each daughter's share
Overall Performance	This question aimed to assess candidates' understanding of shares in inheritance. Most candidates demonstrated a sound knowledge of the shares allocated to the son, daughter, and widow, and applied the correct formulae in their calculations. However, a few candidates showed limited understanding of the share inherited by the widow, indicating a need for greater clarity in this specific area.
Description of Better Responses	In *better responses*, candidates demonstrated a strong understanding of the widow's share by correctly multiplying the total amount of Rs 300,000 by 8 and accurately calculating her portion. They clearly grasped the method for determining the shares of both the son and the daughters, applying the appropriate ratio and inheritance rules. The responses reflected a well-developed conceptual understanding, as candidates not only identified the correct shares but also calculated the share of each daughter with accuracy and confidence.

Images of Better Responses	Widow = $\frac{1}{8} \times x = 300,000$ $x = 300,000 \times 8$ 2400000 The total worth of property is 2,400,000 ii. share of each daughter. (2 Marks) Amount left = 2,400,000 − 300,000 = 2,100,000 Sum of ratios 1:1:2 = 4 $\frac{1}{4} \times 2100000$ Share of each Daughter = 525000 Sheare of two daughter = 1050000
Description of Weaker Responses	In *weaker responses*, candidates demonstrated a reasonable understanding of the widow's share. However, they were unable to accurately calculate the total worth of the property. Many showed limited understanding of the principle that the son's share is double that of the daughter's, which led to incorrect distribution of shares. They were required to multiply 8 by 300,000, however, they mistakenly divided the values instead. This reflected a weak understanding of how to correctly shift an operation across the equality sign. Their responses reflected confusion in applying the correct ratios, and the final calculations often lacked accuracy and consistency.
Images of Weaker Responses	i. total worth of the property. widow $\frac{1}{8}$ 300,000 ÷ 8 = 37500 Shared son = 300,000 ÷ 2 150000 ii. share of each daughter. 37500 + 150000 = 187500 300,000 − 187500 share of each doughter are 112500

Suggestions for improvement (Highlight all that apply)

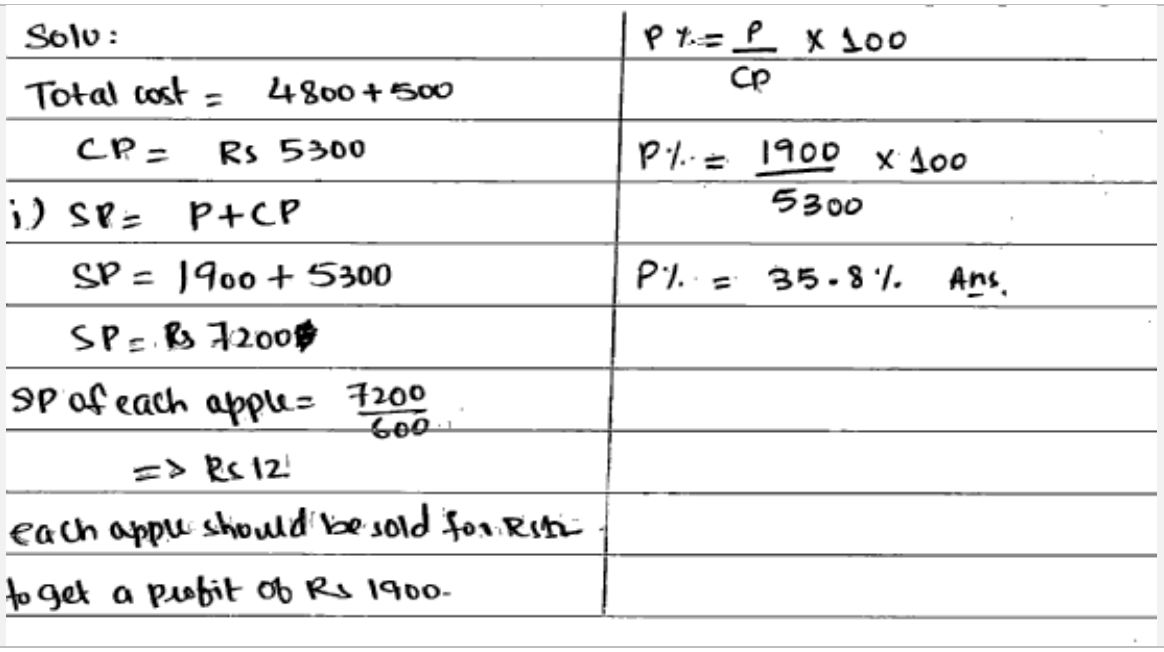
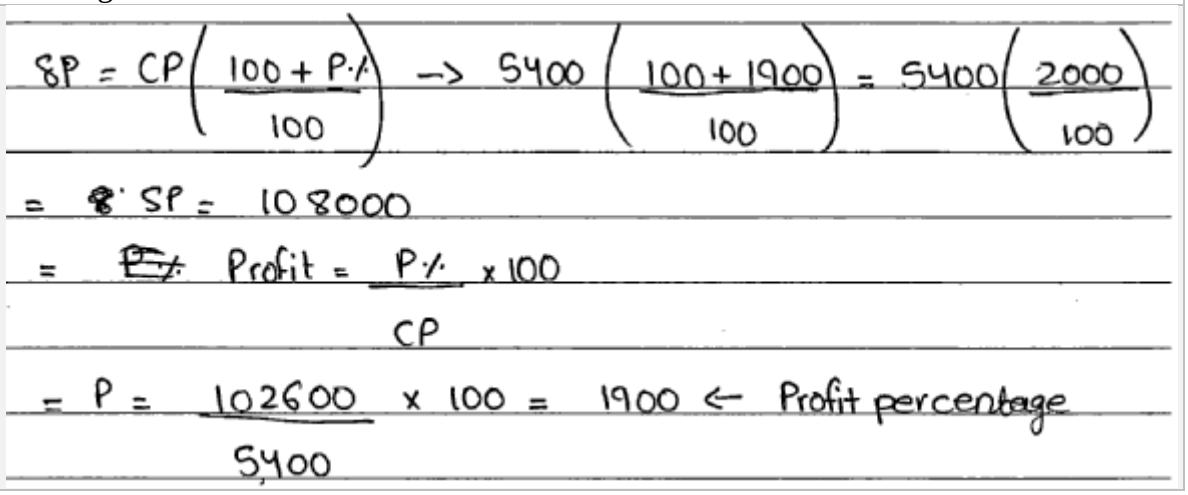
Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies

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Any Additional Suggestion:
Teachers should provide structured opportunities for candidates to engage with a variety of inheritance-based problems that involve both conceptual understanding and ratio-based calculations. The use of step-by-step breakdowns, visual representations (such as bar models or ratio charts), and collaborative problem-solving tasks can help reinforce the principle of proportional distribution.

Question No. 2

Question Text	A fruit vendor bought 600 apples for Rs 4,800. He spent Rs 500 on transportation. Find the price at which he should sell each apple to get a total profit of Rs 1,900. Also, find his profit percentage.
SLO No.	3.3.2
SLO Text	Calculate: a. cost price and selling price; b. profit and profit percentage.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for finding CP 1 mark for finding SP 1 mark for finding SP of each apple 1 mark for finding % age profit
Overall Performance	Overall, a good number of candidates attempted this question successfully, demonstrating the ability to apply the correct formulae for selling price and profit percentage. Most responses reflected a sound understanding of the basic concepts. However, a few candidates struggled with distinguishing between profit and profit percentage, which affected the accuracy of their final answers.
Description of Better Responses	In *better responses*, candidates accurately calculated both the total cost price and the selling price, demonstrating a solid grasp of the underlying concepts. They showed clear understanding in determining the sale price of each apple by dividing 7200 by 600, ensuring precision in their working. Furthermore, the correct profit percentage formula was applied $\%\text{age profit} = \frac{\text{Profit}}{\text{CP}} \times 100$, with appropriate substitution of values and logical progression of steps. The clarity and structure of their working also indicated well-developed problem-solving skills.

Image of Better Response	
Description of Weaker Responses	In *weaker responses*, incorrect values were substituted into the selling price formula. In several cases, the profit amount was mistakenly used in place of the profit percentage, which led to an inaccurate calculation of the selling price. Candidates demonstrated a weak understanding of percentage concepts and struggled to apply the appropriate formulae. They used SP = CP($\frac{100+P\%}{100}$) due to which they got the SP wrong. Moreover, many appeared unable to fully comprehend the requirements of the question, resulting in disorganised working and incorrect final answers.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

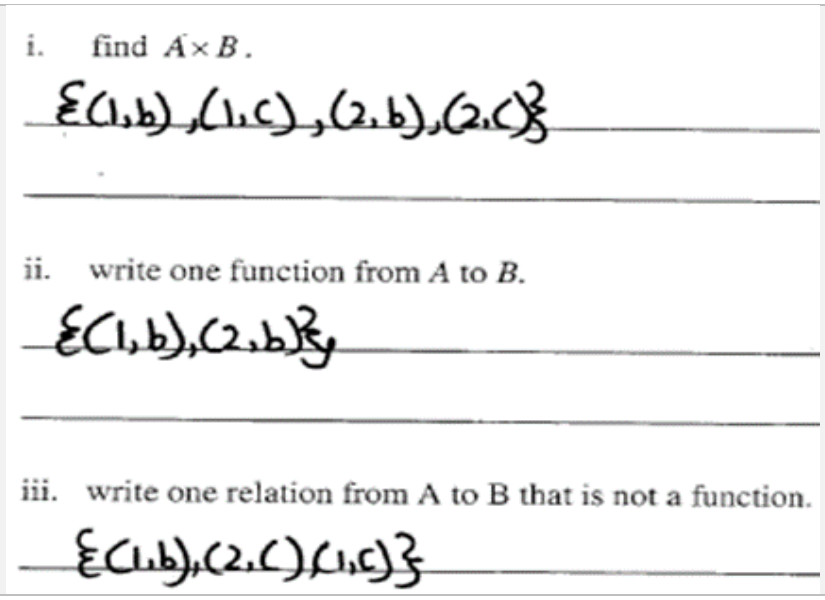
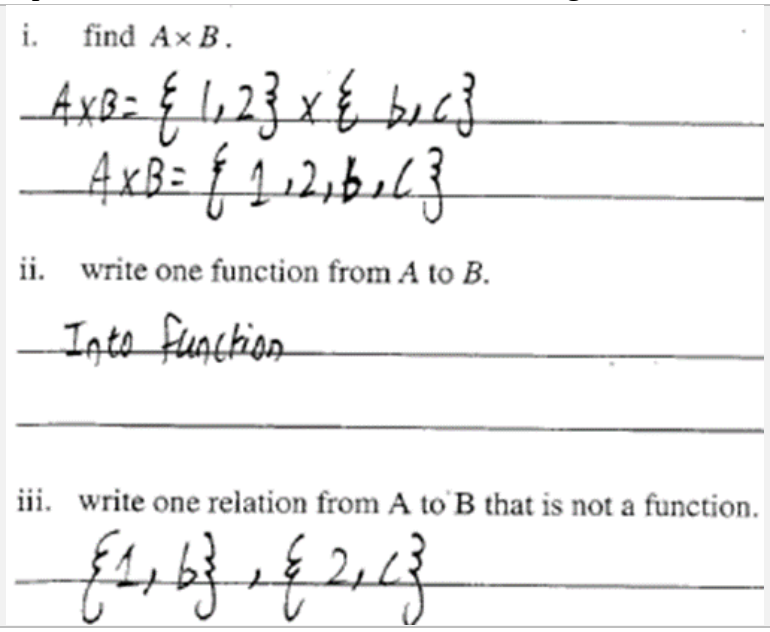
Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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• Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Questioning Technique (Socratic approach) • Practical Demonstration	

Any Additional Suggestion:
Teachers can support students in distinguishing between profit amount and profit percentage through targeted practice and guided exercises. Placing greater emphasis on interpreting key terms within word problems would help develop clarity and accuracy. The use of scaffolded examples and real-life scenarios can further enhance candidates' ability to select and apply the correct formulae, while also improving their overall problem interpretation skills.

Question No. 3

Question Text	If two sets are $A = \{1, 2\}$ and $B = \{b, c\}$, then find $A \times B$. write one function from A to B. write one relation from A to B that is not a function.
SLO No.	4.4.2/ 4.5.2/ 4.6.3
SLO Text	Solve problems related to SLO 4.4.1. Find the domain, co-domain and range of binary relations. Find function from one set to another set.
Max Marks	3
Cognitive Level	U
Checking Hints	1 mark for the Cartesian product (granted only for all correct entries) 1 mark for writing a correct function 1 mark for writing a relation that is not a function
Overall Performance	Most candidates performed well in this question, demonstrating a good understanding of sets, functions, and relations. The majority were able to correctly identify and represent functions using a unique domain and provided accurate descriptions of the given relations. However, a few candidates struggled with the correct notation for subsets or ordered pairs of A×B, particularly with the use of appropriate brackets, which affected the accuracy of their responses.
Description of Better Responses	In *better responses*, candidates accurately listed all the subsets resulting from A×B e.g., $A \times B = \{(1, b), (1, c), (2, b), (2, c)\}$, using correct brackets and appropriate separators to represent the ordered pairs. They demonstrated a clear understanding of the domain and range when identifying the function and showed awareness of the importance of using a unique domain. Additionally, they displayed a sound grasp of the concept of repetition within the domain when writing the relation e.g., $r = \{(1, b), (1, c)\}$, reflecting a well-developed understanding of fundamental set theory and function notation.

Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates struggled to correctly represent the subsets of A×B, often using incorrect brackets and separators for the ordered pairs. Many misinterpreted the question by stating the type of function that could be formed, rather than actually writing the required function. Additionally, instead of presenting the relation, some candidates incorrectly repeated the function, indicating confusion about the distinction between a function and a relation. This suggested a lack of understanding of the rule regarding the repetition of domain elements when defining a relation.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies

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Any Additional Suggestion:		

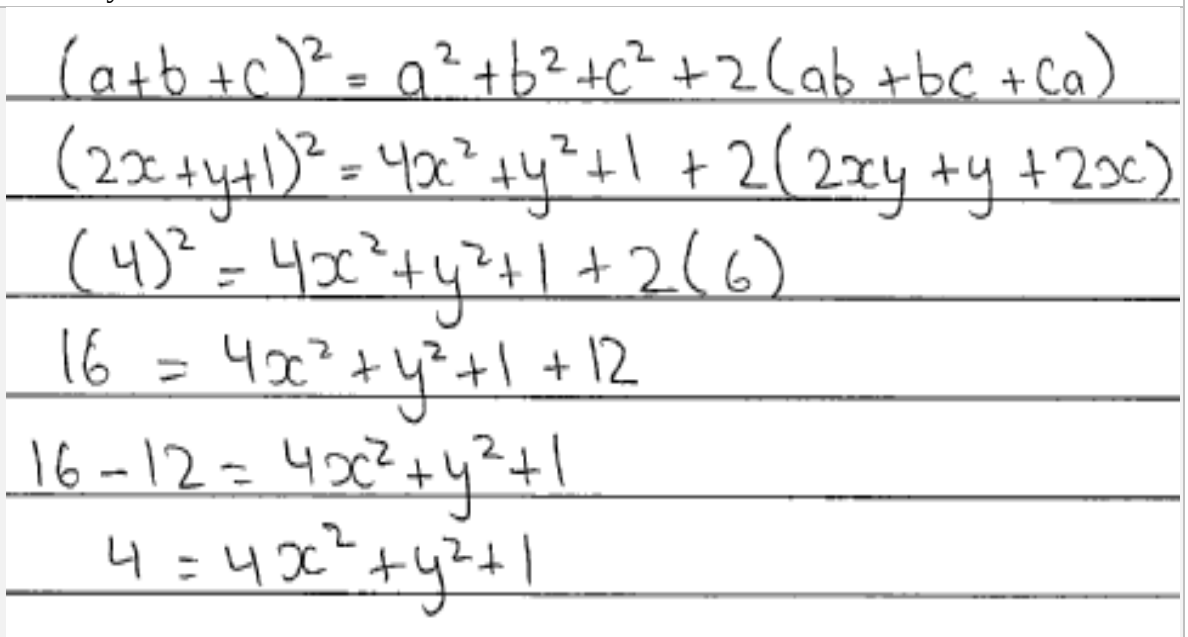
Question No. 4	
Question Text	Applying laws of logarithms, expand the expression $\log\sqrt[3]{\frac{(a+b)^2}{c}}$.
SLO No.	5.5.1
SLO Text	Apply the following laws of logarithm to solve related problems (without using log and antilog tables).
Max Marks	3
Cognitive Level	A
Checking Hints	1 mark for applying first law (power law) correctly 1 mark for applying second law (division law) correctly 1 mark for applying third law (power law) correctly
Overall Performance	Many struggled with the concept of expanding expressions using the laws of logarithms, as well as with converting radical expressions into exponential form. Only a few candidates were able to approach the question successfully through the correct application of logarithmic laws. Overall, the responses indicated that this area requires further reinforcement and conceptual clarity.
Description of Better Responses	In *better responses*, candidates demonstrated a strong understanding of converting radical expressions into exponential form. They correctly applied the power law and quotient law of logarithms to rewrite the given expression, such as converting $log\left\{\frac{(a+b)^2}{c}\right\}^{\frac{1}{3}}$ as $\frac{1}{3}\{log(a+b)^2 - \log c\}$. These responses reflected a clear conceptual understanding and confident application of logarithmic properties.

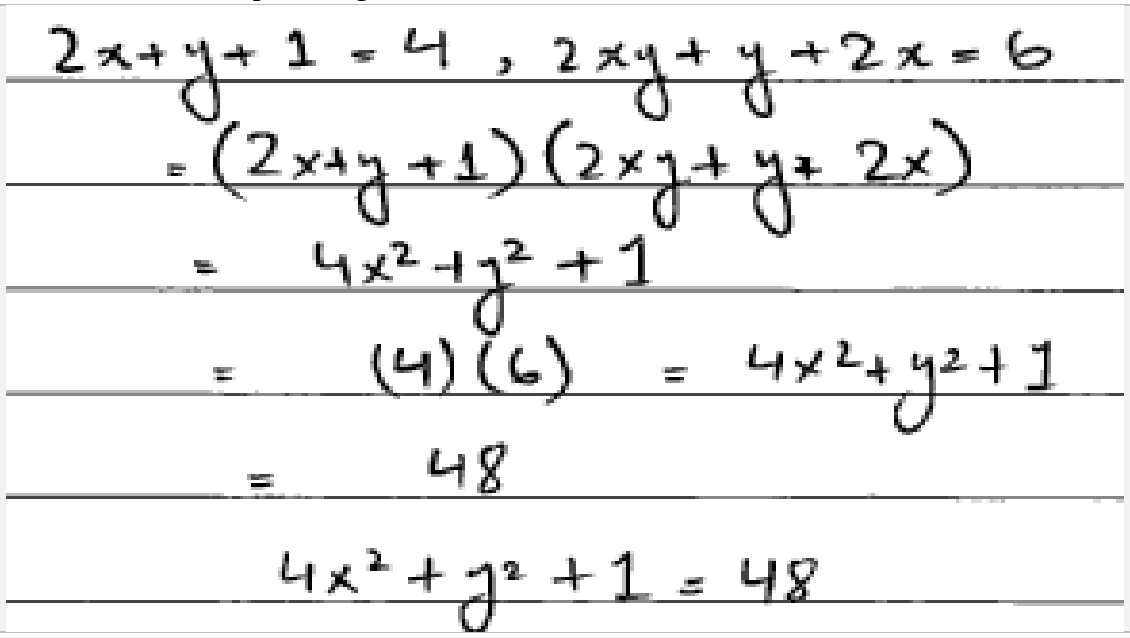
Image of Better Response	$\log\left(\frac{(a+b)^2}{c}\right)^{1/3}$ $\frac{1}{3}\log(a+b)^2 - \frac{1}{3}\log c$ $\frac{2}{3}\log(a+b) - \frac{1}{3}\log c$
Description of Weaker Responses	In *weaker responses*, candidates demonstrated a weaker understanding of the laws of logarithms. Many candidates incorrectly converted the exponent into the base of the logarithmic expression, which reflected a fundamental confusion between the laws of exponents and the laws of logarithms. Rather than applying the appropriate logarithmic rules such as the power $\log_a(m)^n = n\log_a m$ or quotient laws $\log_a\left(\frac{m}{n}\right) = \log_a m - \log_a n$, candidates began using exponent rules, leading to incorrect and illogical expressions.
Image of Weaker Response	$= \log_3(a+b)^2 - \log_3(c)$ $= \log_3(a+b)^2 \times (c)$ $= \log_3(a^2 * b^2 \times c)$ $\log_3(a^2 \cdot b^2 \cdot c)$ $\log(a^2 \cdot b^2 \cdot c^{\#})^3$ $\log(a^{2-3} b^{2-3} c^{-3})$ $\log(a^{-1} b^{-1} c^{-3})$ $\log = a^{-1} b^{-1} c^{-3}$ $\log \frac{a^{-1} b^{-1}}{c^{-3}}$

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:		

Teachers should adopt a dual-strategy approach that explicitly contrasts the laws of exponents with those of logarithms, using side-by-side examples to highlight their distinct roles and applications. Incorporating concept mapping and structured comparison tasks can help candidates internalise these differences.

Question No. 5	
Question Text	For the expressions $2x+y+1=4$ and $2xy+y+2x=6$, find the value of $4x^2+y^2+1$.
SLO No.	6.2.2
SLO Text	Find the values of $a^2 + b^2 + c^2$, a + b + c and ab+ bc+ ca, when values of any two of these are given using the formula $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for squaring both sides 1 mark for correct expansion 1 mark for substitution and multiplying $2xy+y+2x$ by 2 1 mark for writing $4x^2+y^2+1=4$
Overall Performance	Although the majority of candidates were unable to fully comprehend the question and struggled to apply the correct identity to expand the given expression, very few candidates were able to select the appropriate identity and successfully obtained the correct answer. Overall, the responses indicated that many candidates required a stronger understanding of algebraic identities and their application in expansion problems.
Description of Better Responses	In *better responses*, candidates demonstrated a strong understanding of the identity $(a+b+c)^2=a^2+b^2+c^2+2ab+2bc+2ca$ and arranged the terms correctly. They accurately substituted the given values and carried out the simplification in a systematic and logical manner. Their clear application of the identity and careful handling of each step reflected a solid grasp of algebraic expansion and contributed to arriving at the correct final answer $4x^2+y^2+1=4$. These responses showed both conceptual clarity and procedural accuracy.
Image of Better Response	

Description of Weaker Responses	In *weaker responses*, candidates were unable to comprehend the requirements of the question and struggled to recall the appropriate identity to be applied. Rather than using the correct identity $(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca$, many candidates attempted to find the product of the given values directly $(2x+y+1)(2xy+y+2x)$. This indicated a lack of familiarity with algebraic identities and a limited understanding of expansion techniques. The responses reflected the need for further reinforcement of foundational concepts in algebra.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestion:

Teachers should give greater importance to explicit instruction and repeated practice on core algebraic identities, particularly those involving the square of a trinomial such as $(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca$. Using pattern recognition strategies and structured expansion tasks, students can be guided to see the connection between the identity and the question context.

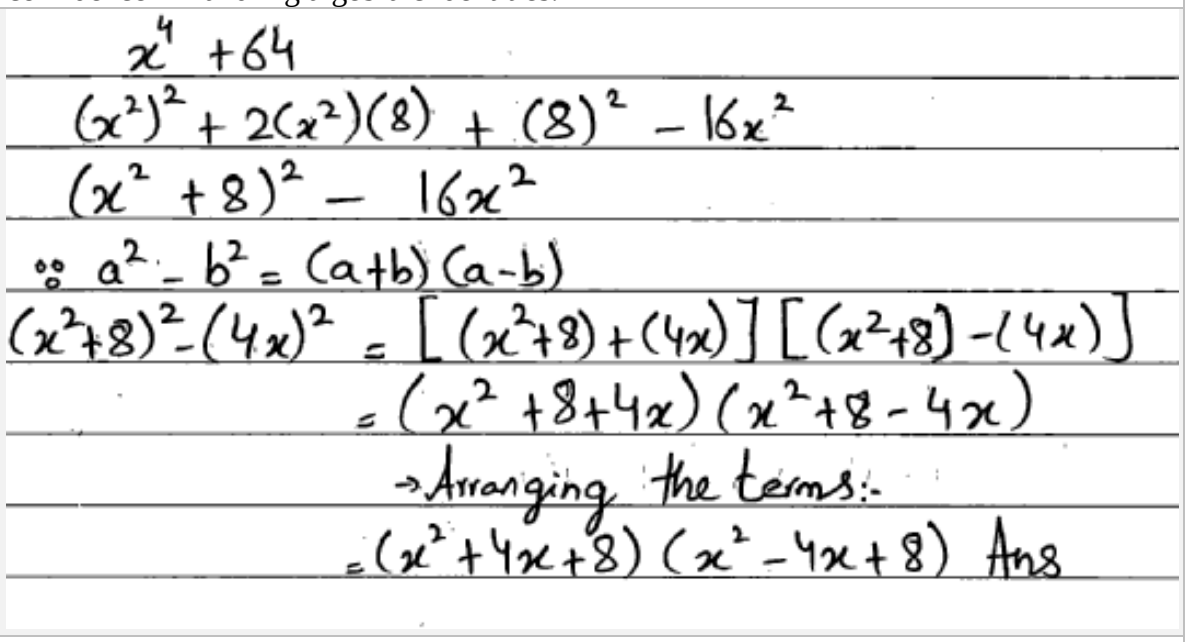
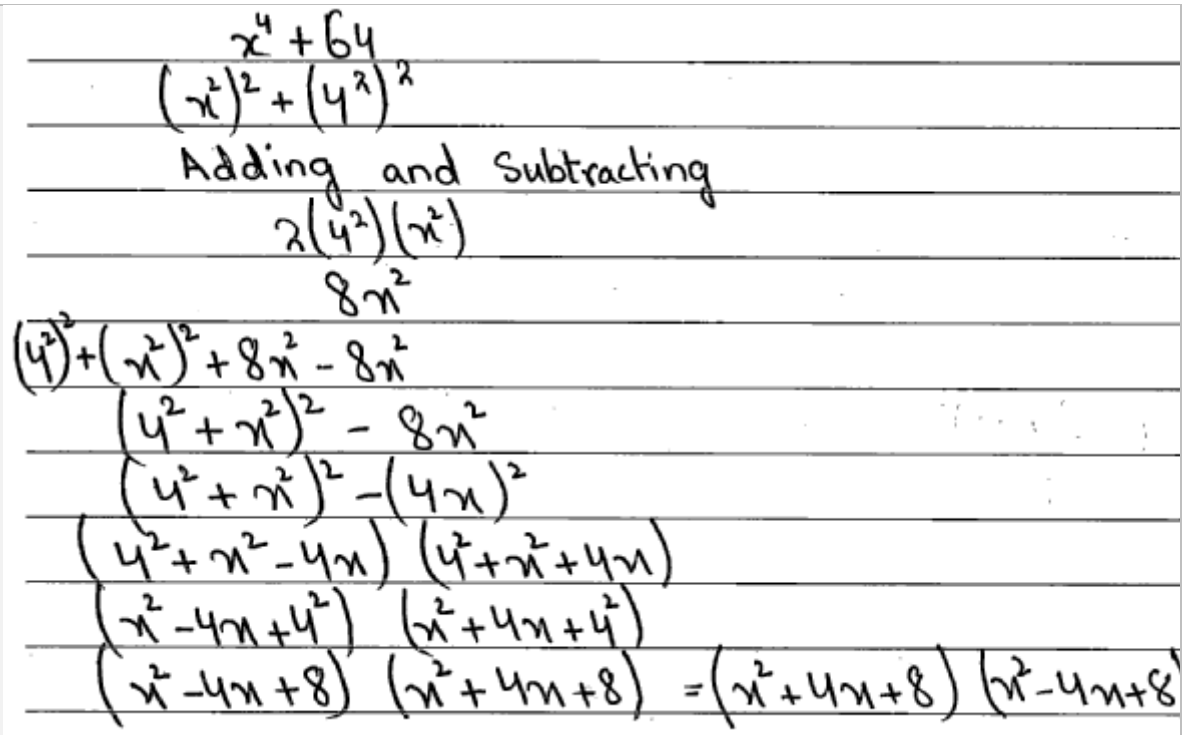
Question No. 6a Candidates were given the choice to attempt any ONE out of the two questions: 6a and 6b.	
Question Text	Prove that the factors of x^4+64 are (x^2+4x+8) and (x^2-4x+8).
SLO No.	7.1.1f
SLO Text	Factorise the expression of the following types.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for writing the squares $(x^2)^2+(8)^2$ 1 mark for identifying the right terms to be added and subtracted as $\pm 16x^2$ 1 mark for writing in the form $(x^2+8)^2-16x^2$ 1 mark for writing $(x^2+4x+8)(x^2-4x+8)$
Overall Performance	This was a choice-based question, and only a few candidates opted to attempt this part. Among those who did, better-performing candidates effectively used both factorisation and expansion methods to arrive at the required result. However, the overall performance on this question reflected a general lack of understanding in identifying the correct method of factorisation. This highlights the need for greater conceptual clarity and practice in choosing and applying appropriate algebraic methods.
Description of Better Responses	In *better responses*, candidates demonstrated a strong understanding of identifying the appropriate method of factorisation, namely completing the square. They correctly added and subtracted the term $16x^2$ and proceeded accurately by applying the identity a^2-b^2. This allowed them to successfully factorise the expression into $(x^2+4x+8)(x^2-4x+8)$. Their working was systematic and well-reasoned, reflecting both conceptual clarity and confidence in handling algebraic identities.
Image of Better Response	
Description of Weaker Responses	In *weaker responses*, candidates demonstrated limited understanding of finding the square of terms. Several candidates incorrectly added $8x^2$ instead of the required $16x^2$, which led to errors in the subsequent steps. Although some attempted to use the correct method of completing the square, their weaker grasp of squaring expressions prevented them from arriving at the correct answer. These responses reflected a need for greater conceptual clarity and more practice in applying algebraic identities accurately.

Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

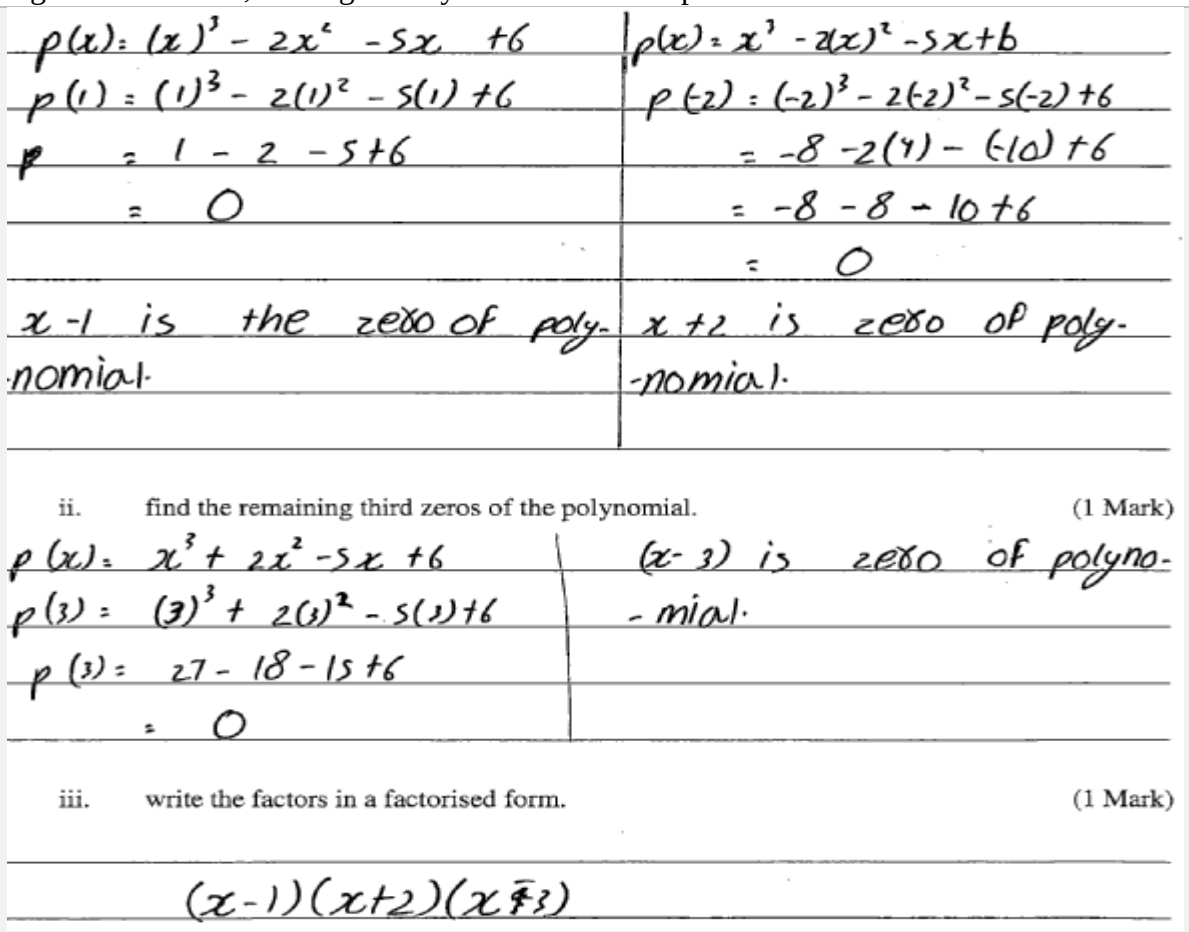
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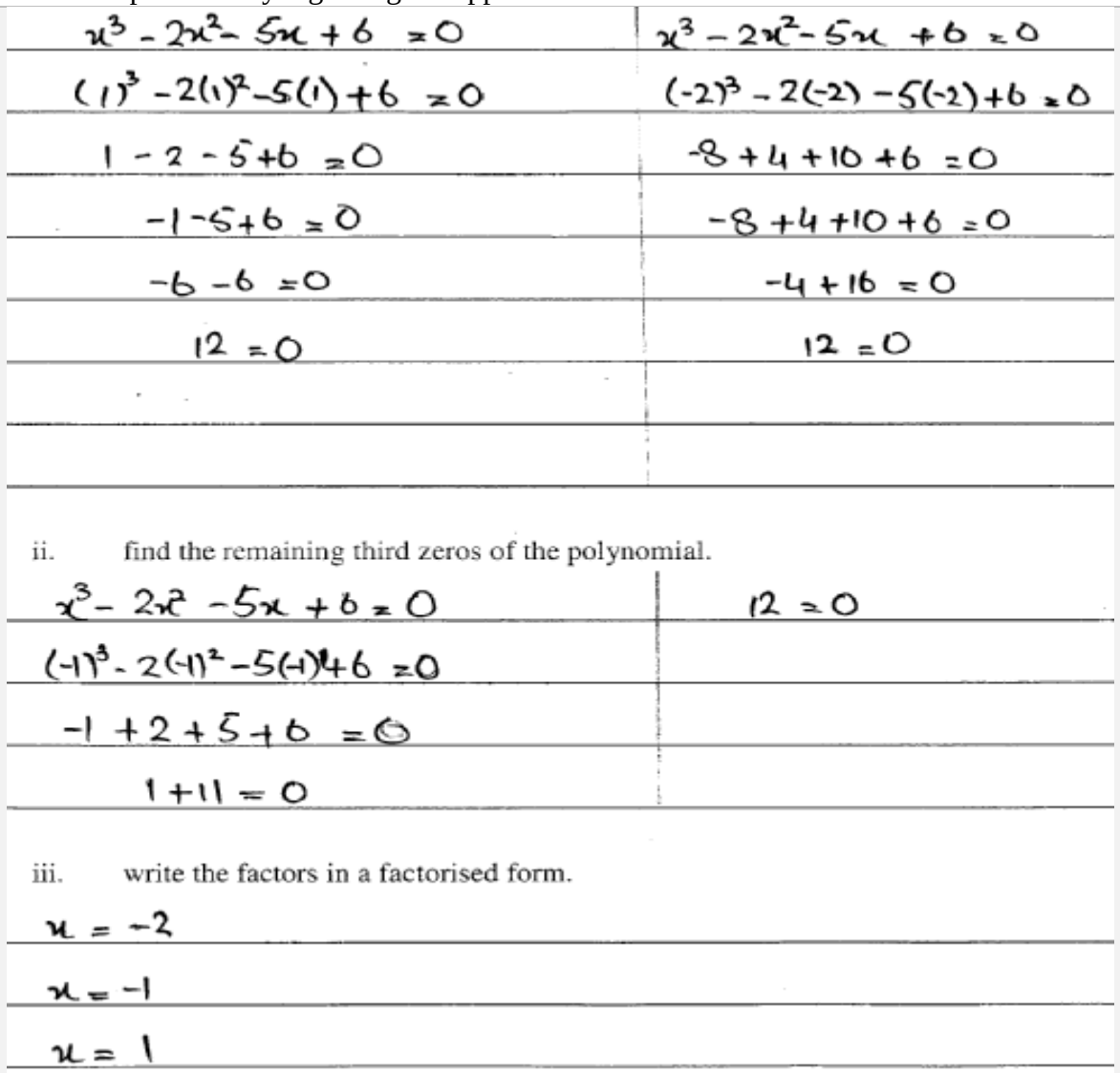
Any Additional Suggestion:
Teachers incorporate key algebraic identities, particularly those involving the square of binomials, using concrete examples and visual representations (such as algebra tiles or area models) to reinforce the structure of expressions like $(a\pm b)^2$. Providing step-by-step guided practice that explicitly breaks down the squaring process can help candidates internalise the correct expansion.

Question No. 6b

Candidates were given the choice to attempt any ONE out of the two questions: 6a and 6b.

Question Text	For the polynomial x^3-2x^2-5x+6, i. Show that 1 and -2 are zeros of the given polynomial. ii. Find the remaining third zeros of the polynomial. iii. Write the factors in a factorised form.

SLO No.	7.3.1 and 7.2.3
SLO Text	Solve problems based on the concept of zeros of a polynomial Apply factor theorem to factorise a cubic polynomial.
Max Marks	4
Cognitive Level	A
Checking Hints	i. 1 mark for proving 1 as one of the zeros of the polynomial ii. 1 mark for proving -2 as one of the zeros of the polynomial iii. 1 mark for finding the remaining zeros of the polynomial iv. 1 mark for writing the factors of the polynomial (Mark will be given on any one correct factor).
Overall Performance	This was a choice-based question, and the majority of candidates attempted this part. Most candidates demonstrated a sound understanding of the factor theorem and were able to correctly identify the factors of the given expression. However, a few candidates struggled with the simplification steps, which affected the accuracy of their final answers. Additionally, some candidates showed limited understanding in writing the factors in the correct algebraic form. Overall, while the general performance was satisfactory, greater attention to detail and procedural accuracy is still needed.
Description of Better Responses	In *better responses*, candidates demonstrated a clear understanding of the factor theorem and successfully verified the zeroes of the given polynomial $p(x) = 1 - 2 - 5 + 6 = 0$. They proceeded methodically to determine the third factor and expressed all three factors accurately and in the correct algebraic form $(x - 3)(x + 2)(x - 1)$. Their responses reflected strong conceptual understanding, precise application of the theorem, and attention to algebraic structure, leading to fully correct and well-presented solutions.
Images of Better Responses	

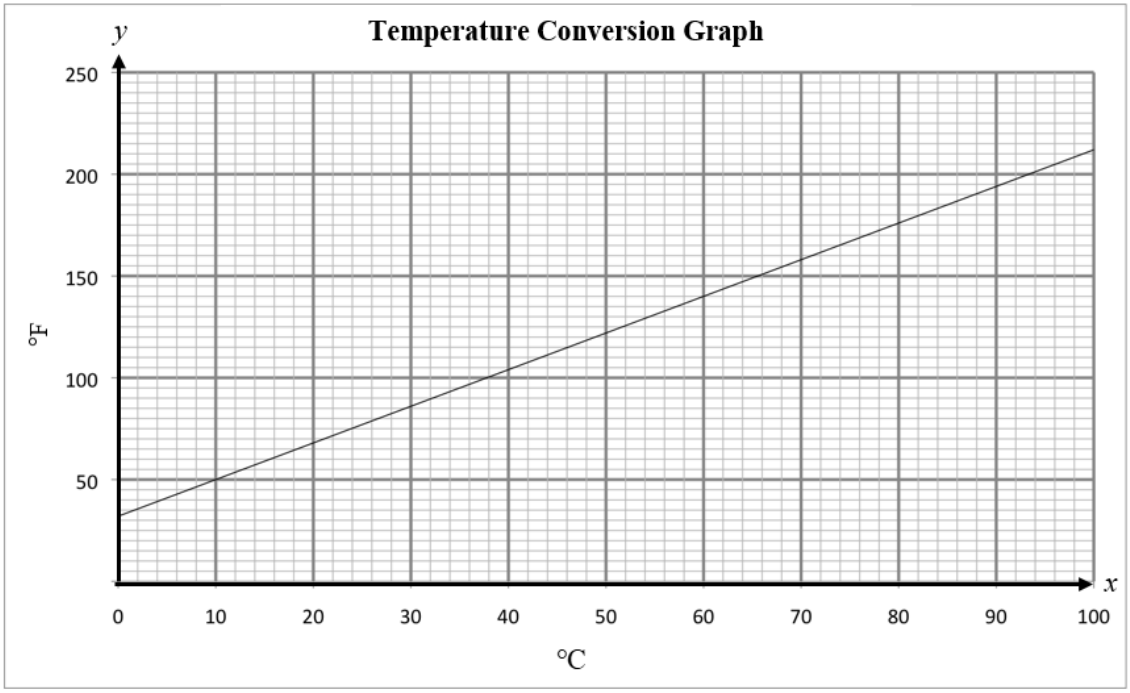
Description of Weaker Responses	In *weaker responses*, candidates were able to substitute the correct zeroes of the polynomial. However, errors in simplification e.g., writing -6 at the place of +6 led to incorrect results. Furthermore, they were unable to determine the third factor and demonstrated limited understanding in expressing the factors in the appropriate algebraic form, they wrote x = -1 or x = 1 instead of factor form. These responses indicated gaps in both procedural accuracy and conceptual clarity regarding the application of the factor theorem.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

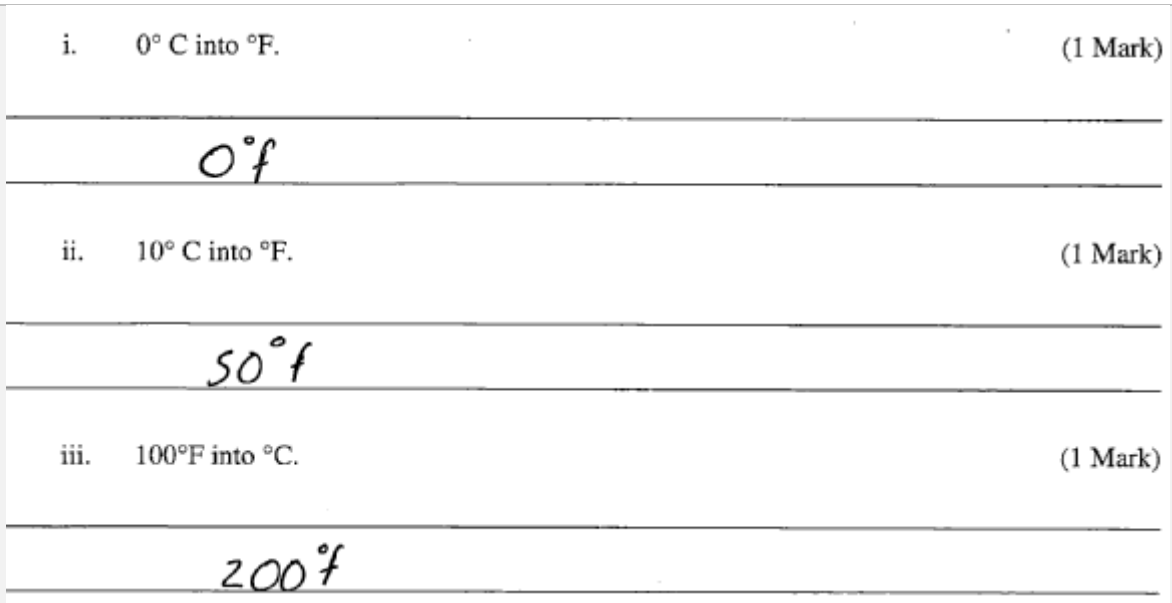
Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

• Utilise the resource guide for additional materials	• Questioning Technique (Socratic approach) • Practical Demonstration	

Any Additional Suggestion:
To strengthen candidates' procedural accuracy and conceptual understanding, teachers should incorporate structured factorisation exercises that focus on linking known zeroes to their corresponding linear factors and systematically deriving the complete factorised form. Emphasis should be placed on verifying factors through substitution and reconstructing the polynomial via multiplication to build confidence in their results.

Question No. 7	
Question Text	The given graph shows relationship between the temperature measured in degree °F and degree °C scales.  Using the information in the graph, convert i. 0° C into °F. ii. 10° C into °F. iii. 100° F into °C.
SLO No.	8.2.2
SLO Text	Use graph for conversions of: c. degrees Celsius and degrees Fahrenheit.
Max Marks	3
Cognitive Level	A

Checking Hints	i. 0° C = 32 °F (or any value greater than 30 and less than 35) ii. 10° C = 50°F iii. 100°F into 38°C (up to two degrees less or two degrees more will be acceptable)
Overall Performance	A good number of candidates performed well in this question by correctly reading the graph and accurately reading the scale. They also applied the relevant formula appropriately $°F = \left(\frac{9}{5} \times °C\right) + 32$, demonstrating a clear understanding of the required concepts. However, a few candidates struggled to read the graph correctly, which led to incorrect converted values. Overall, the performance indicated satisfactory understanding, though some candidates would benefit from further practice in graph reading
Description of Better Responses	In *better responses*, candidates demonstrated a strong understanding of reading conversion graphs. They showed clear awareness of the scale used on both the x-axis and y-axis and accurately identified the converted values from the graph e.g., 0° C = 32 °F. Their answers included correct units and were presented with clarity and precision. These responses reflected a solid grasp of graphical reading and careful attention to detail.
Image of Better Response	i. 0° C into °F. (1 Mark) 0°C = 30 °F ii. 10° C into °F. (1 Mark) 10°C into 50 °F iii. 100°F into °C. (1 Mark) 100°F into 38°C
Description of Weaker Responses	In *weaker responses*, candidates demonstrated a limited understanding of reading scales and extracting accurate values from the graph. Some candidates attempted to apply the relevant formula $°F = \left(\frac{9}{5} \times °C\right) + 32$. This indicated a lack of fluency in both interpreting graphical data and performing accurate calculations, highlighting the need for further practice in these foundational skills.

Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:		

Question No. 8a	
Candidates were given the choice to attempt any ONE out of the two questions: 8a and 8b.	
Question Text	If $A=\begin{bmatrix}1 & -1\\ 0 & 2\end{bmatrix}$ and $B=\begin{bmatrix}2 & 1\\ 0 & 1\end{bmatrix}$, then prove that $(AB)^t = B^tA^t$.
SLO No.	9.4.7
SLO Text	Verify with the help of examples that $(AB)^t = B^tA^t$.
Max Marks	3
Cognitive Level	A

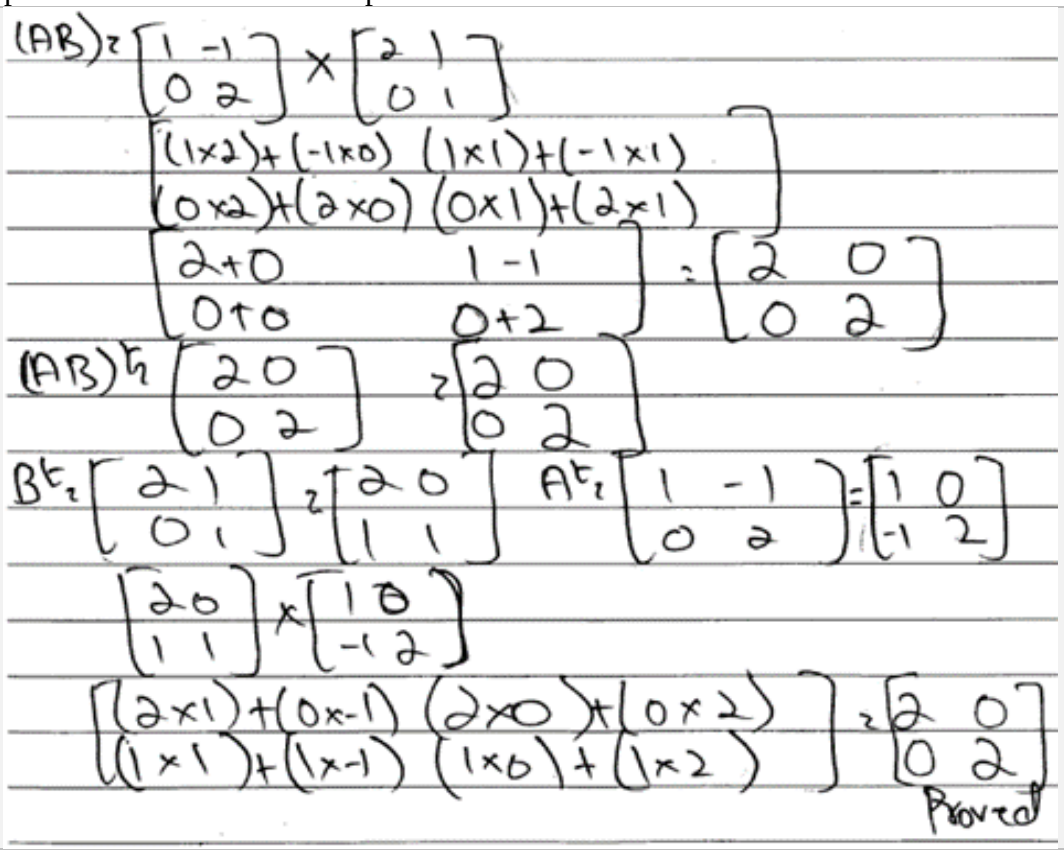
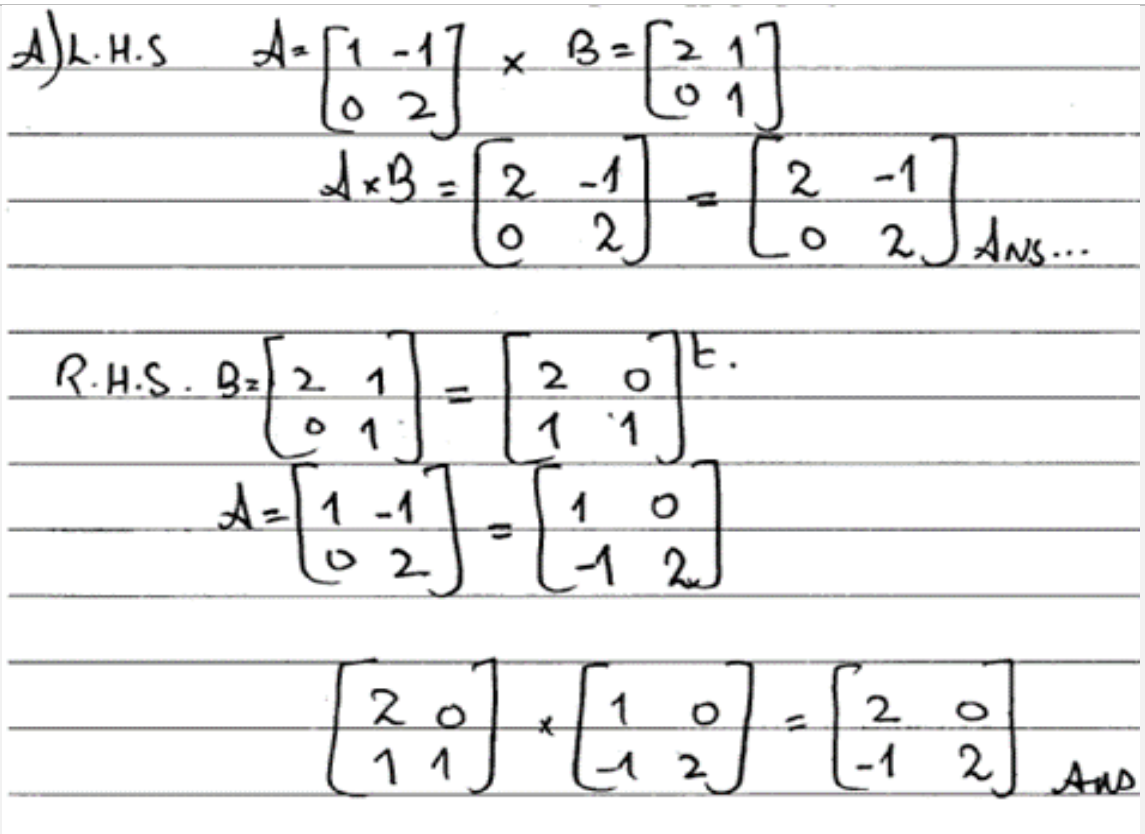
Checking Hints	1 mark for correct A^t and B^t (mark would be granted for even one of these is correct) 1 mark for correct matrix multiplication AB 1 mark for correct matrix multiplication $B^t\ A^t$
Overall Performance	The candidates were given a choice to attempt either Question 8a or 8b. The majority opted for 8a. Among those who attempted this part, majority demonstrated a good understanding of the concepts involved in proving the identity $(AB)^t=B^tA^t$. However,errors in matrix multiplication, the incorrect use of signs during the process, and mistakes in taking the transpose of the resulting matrix. Overall, while conceptual understanding appeared sound, procedural accuracy remained an area needing improvement.
Description of Better Responses	*Better responses* showed that candidates clearly demonstrated all the necessary steps to prove the identity $(AB)^t=B^tA^t$. They accurately carried out the multiplication of matrices A and B and correctly determined the transpose of $(AB)^t$, as well as A^t and B^t. Their working was logical and systematic, reflecting a strong conceptual understanding and sound procedural skills in matrix operations.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* showed that candidates faced difficulty in multiplying the two matrices correctly. Although they were able to take the transpose of the individual matrices, i.e., A^t and B^t, accurately, their limited understanding of matrix multiplication prevented them from successfully proving the identity $(AB)^t = B^tA^t$. These responses reflected a need for further reinforcement of the rules and procedures involved in matrix operations.

Image of Weaker Response	

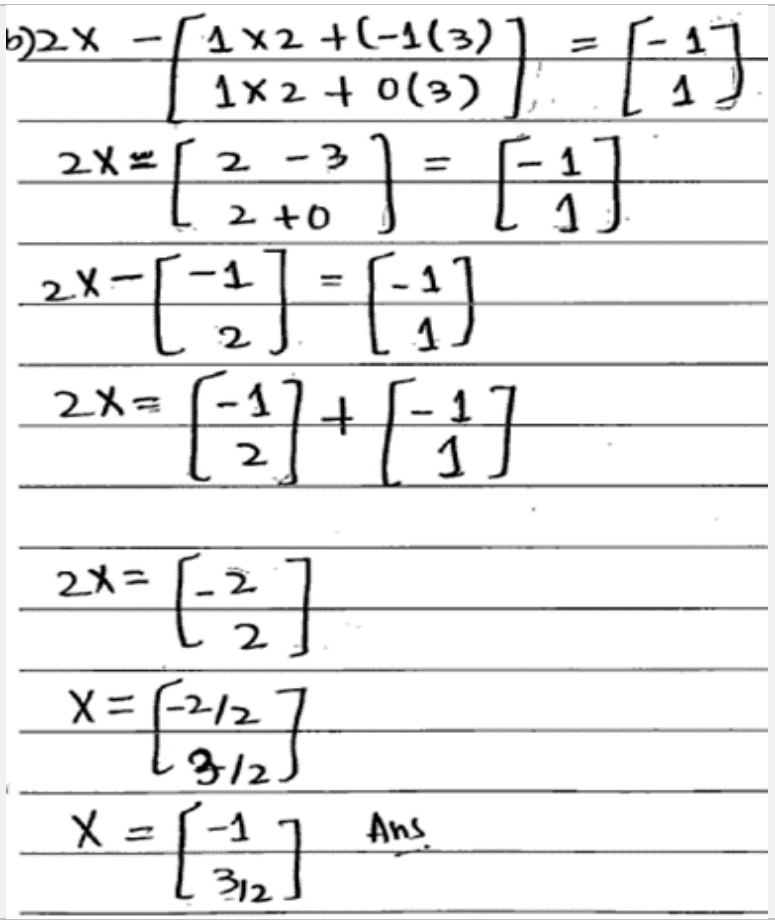
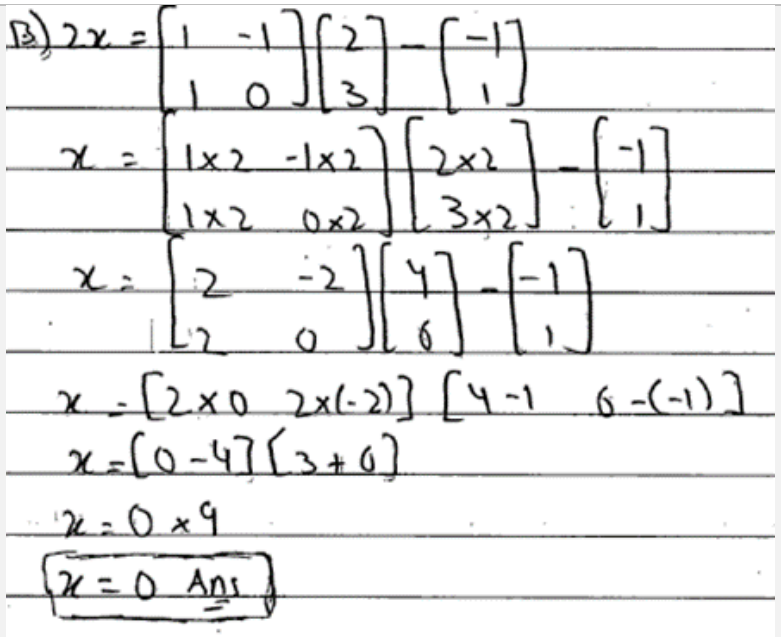
Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Preferred Pedagogy Used for this SLO**	**Assessment Strategies**
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Any Additional Suggestion:

Teachers should provide targeted practice on the order of multiplication in matrices, emphasising that matrix multiplication is not commutative. Incorporating step-by-step guided examples comparing $(AB)^t$ and B^tA^t side by side can help students visually and logically grasp the reversal in order of factors during transposition.

Question No. 8b	
Candidates were given the choice to attempt any ONE out of the two questions: 8a and 8b.	
Question Text	Solve the given matrix equation to find the unknown matrix X. $2X - \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$
SLO No.	9.6.1
SLO Text	Solve matrix equations.
Max Marks	3
Cognitive Level	A
Checking Hints	1 mark for correct multiplication i.e. $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$ 1 mark for adding on the RHS as $\begin{bmatrix} -1-1 \\ 1+2 \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ 1 mark for dividing the matrix obtained on the RHS by 2
Overall Performance	The majority of those who attempted it were able to correctly find the value of 'X'. However, some candidates demonstrated a lack of basic understanding in solving equations and encountered difficulty in arranging the known and unknown terms on either side of the equation. Those who were unable to arrive at the correct answer appeared to struggle with the concept of matrix multiplication. Common issues included sign errors and difficulty in eliminating the coefficient of X, which hindered their ability to achieve the desired result.
Description of Better Responses	*Better responses* showed that candidates were able to perform all the necessary steps required to solve the given equation accurately. They correctly carried out the matrix multiplication and incorporated the appropriate signs while adding the matrices. Finally, they successfully divided the resulting matrix by 2 to determine the value of 'X'. These responses reflected a clear understanding of matrix operations and a systematic approach to problem-solving.

Image of Better Response	
Description of Weaker Responses	*Weaker responses* showed that candidates faced difficulty with matrix multiplication. Furthermore, they struggled to isolate the variable 'X' in the equation 2X−[A][B]=[C], which led to incorrect or incomplete solutions. Their inability to manipulate the equation algebraically and apply appropriate matrix operations prevented them from determining the correct value of the unknown. These responses indicated the need for further reinforcement of foundational concepts in matrix algebra.
Image of Weaker Response	

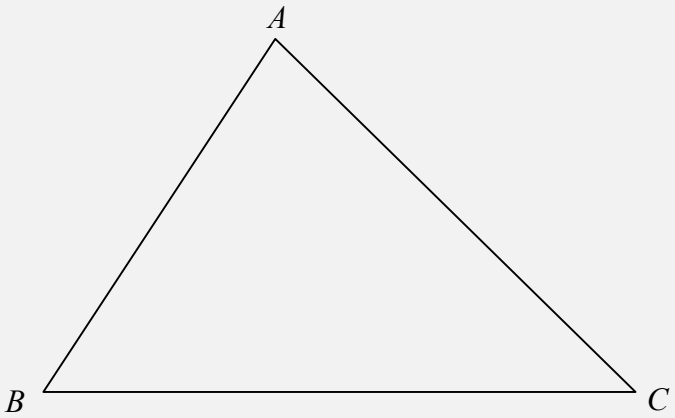
Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestion:

Use Scaffolding with Worked Examples: Begin with simple matrix multiplication using 2×2 matrices before progressing to more complex ones. Provide step-by-step annotated examples on how to multiply matrices and isolate a matrix variable.

Incorporate Visual Aids and Flowcharts: Use flowcharts or procedural maps to help students understand the sequence of operations—first multiply, then isolate, then divide (if applicable).

Question No. 9	
Question Text	The given diagram shows a triangle ABC.  Show that the angle bisectors of the given triangle are concurrent. Also, mention the point of concurrency.
SLO No.	10.1.3a
SLO Text	Verify, for a given triangle, the concurrency of: a. angle bisectors.
Max Marks	3
Cognitive Level	A
Checking Hints	1 mark for each angle bisector 1 mark for the second angle bisector 1 mark for identifying the point of concurrency.

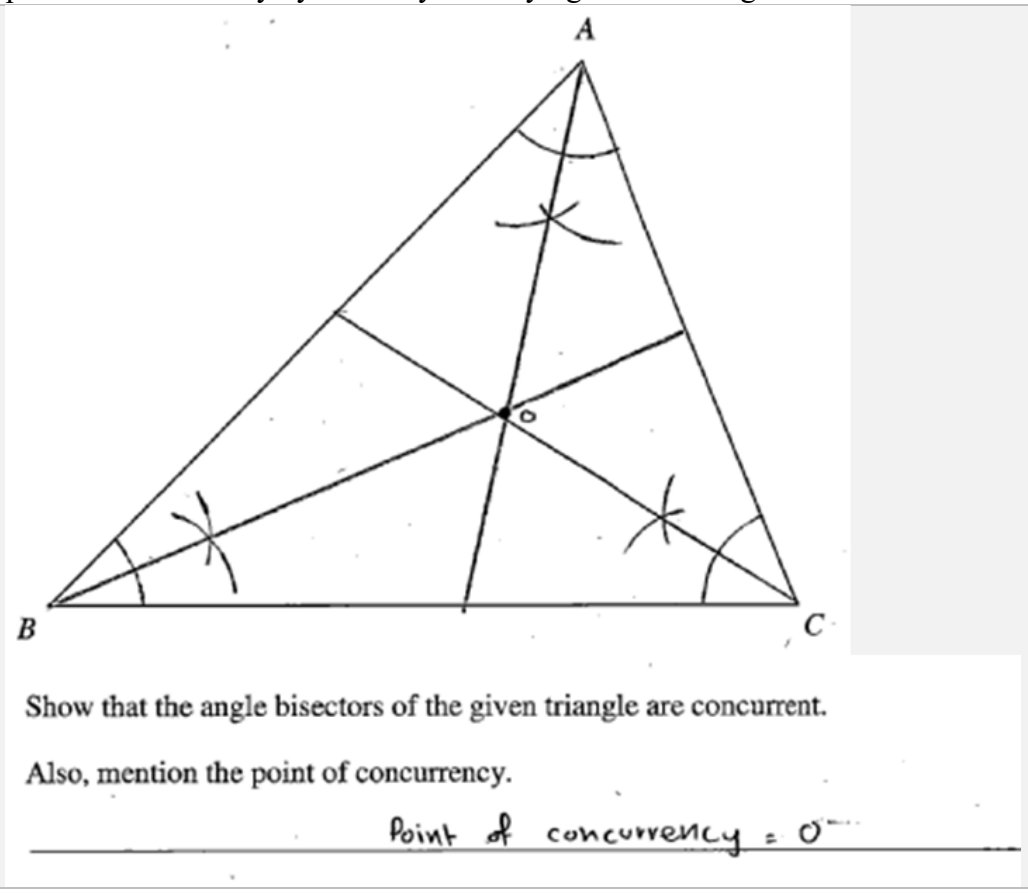
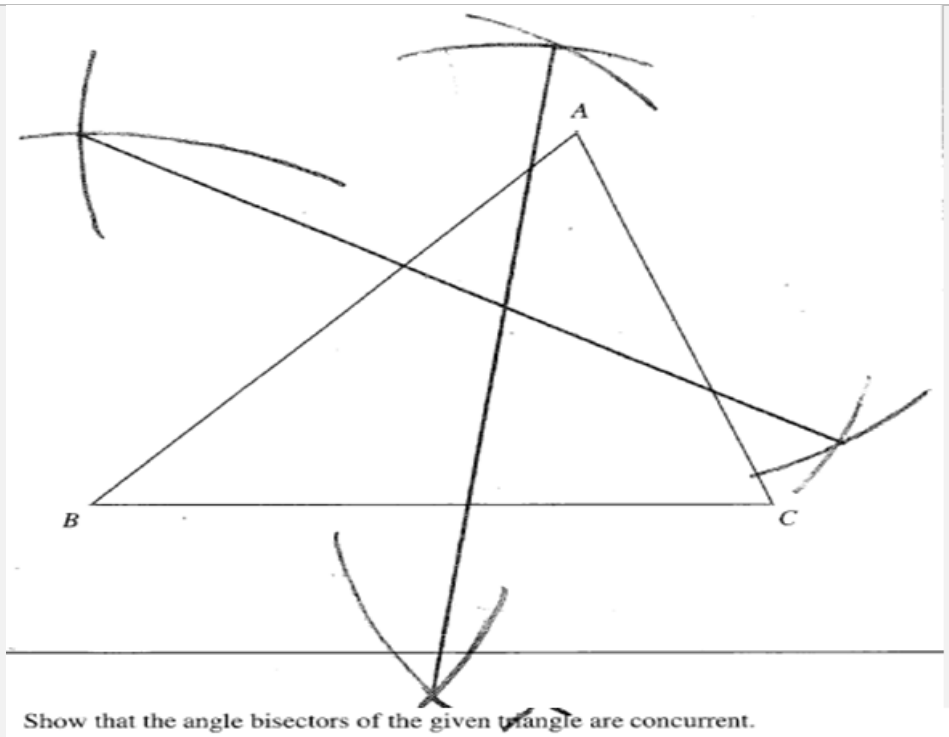
Overall Performance	Most candidates were able to construct the angle bisectors accurately; however, very few labelled them. Those who were unable to construct the angle bisectors appeared to be confused, as many had drawn side bisectors instead. A number of candidates used incorrect arcs or lines, and several constructions included freehand-drawn lines and arcs, which compromised precision. While most candidates successfully located the point of concurrency, only a few took the additional step to label it correctly.
Description of Better Responses	*Better responses* showed that candidates had a clear understanding of how to construct angle bisectors. They accurately drew all the necessary arcs and lines to bisect each angle of the given triangle. Furthermore, they demonstrated a sound understanding of the concept of the point of concurrency by correctly identifying and labelling it with the variable 'O'.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* showed that candidates struggled with the concept of angle bisectors, often drawing incorrect arcs and lines. As a result of these inaccuracies, they were unable to identify or locate the point of concurrency correctly. The lack of precision in construction and confusion between angle and side bisectors reflected limited understanding of the geometric principles involved.

Image of Weaker Response	 Show that the angle bisectors of the given triangle are concurrent. Also, mention the point of concurrency. Point C and AB. Point A and BC.

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

- **Provide Guided Practice Worksheets:** Design worksheets that break down each part of the construction with prompts for students to complete each step before moving on.

- **Peer Review Sessions:** Organise peer-check activities where students compare their constructions and provide feedback using a checklist (e.g., arcs drawn? lines labelled? point of concurrency marked?).

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.

Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.

Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.

Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.

Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources

Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.

Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share

Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.

Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.

Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.

Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

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